Selective Adversarial Adaptation-Based Cross-Scene Change Detection Framework in Remote Sensing Images

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***Abstract*— Supervised change detection methods always face a big challenge that the current scene (target domain) is fully unlabeled. In remote sensing, it is common that we have sufficient labels in another scene (source domain) with a different but related data distribution. In this article, we try to detect changes in the target domain with the help of the prior knowledge learned from multiple source domains. To achieve this goal, we propose a change detection framework based on selective adversarial adaptation. The adaptation between multisource and target domains is fulfilled by two domain discriminators. First, the first domain discriminator regards each scene as an individual domain and is designed for identifying the domain to which each input sample belongs. According to the output of the first domain discriminator, a subset of important samples is selected from multisource domains to train a deep neural network (DNN)-based change detection model. As a result, not only the positive transfer is enhanced but also the negative transfer is alleviated. Second, as for the second domain discriminator, all the selected samples are thought from one domain. Adversarial learning is introduced to align the distributions of the selected source samples and the target ones. Consequently, it further adapts the knowledge of change from the source domain to the target one. At the fine-tuning stage, target samples with reliable labels and the selected source ones are used to jointly fine-tune the change detection model. As the target domain is fully unlabeled, homogeneity- and boundary-based strategies are exploited to make the pseudolabels from a preclassification map reliable. The proposed method is evaluated on three SAR and two optical data sets, and the experimental results have demonstrated its effectiveness and superiority.**

***Index Terms*— Adaptation, adversarial learning, change detection, deep neural networks (DNNs), domain discriminator, remote sensing images.**

Manuscript received February 26, 2020; accepted May 28, 2020. Date of publication June 19, 2020; date of current version February 25, 2021. This work was supported in part by the National Natural Science Foundation of China under Grant U1701267, in part by the State Key Program of National Natural Science of China under Grant 61836009, in part by the Major Research Plan of the National Natural Science Foundation of China under Grant 91438201, and in part by the Foundation for Innovative Research Groups of the National Natural Science Foundation of China under Grant 61621005. *(Corresponding author: Licheng Jiao.)*

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Digital Object Identifier 10.1109/TGRS.2020.3001584

I. Introduction

BY ANALYZING two remote sensing images acquired over the same geographical area at different times, change detection aims at identifying changes that might have occurred on Earth's surface. It plays an important role in numerous fields of applications [1], [2], including land use monitoring, urban planning, disaster evaluation, and environmental investigation. Synthetic aperture radar (SAR) images are commonly used in change detection studies [3], [4] due to the independence of atmospheric and solar illumination conditions. Optical images are also alternatives besides SAR images since they are cheap. Due to the development of satellite missions, an increasing number of remote sensing images are available to help monitor dynamic changes of the land cover. It should be noted that geometric coregistration and radiometric correction are essential to make two input SAR images comparable before change detection.

Over the past decades, various methods are developed to detect changes in remote sensing images [5]. In general, there are three necessary parts in these methods, including removing noise [6], extracting features for comparison, and distinguishing between the changed and unchanged classes. For multispectral images, the difference image (DI) is represented by the spectral change vector (SCV) to extract the change information. Distinguishing between changed and unchanged classes is always thought as an image segmentation or classification problem. From this perspective, many researchers focus their attention on change vector analysis (CVA) [7], [8], statistical modeling [9], [10], clustering approaches [11]–[13], decision tree [14], active contour model [15], [16], and support vector machines (SVMs) [2], [17]. In recent years, deep neural networks (DNNs) have obtained an increasing interest in many disciplines. Various deep architectures were developed and applied in change detection of remote sensing images, such as deep belief networks (DBNs) [18], [19], stacked autoencoders (SAEs) [3], [20], deep siamese networks [21], [22], U-net [23]–[25], VGG-16 network [26], [27], and generative adversarial networks (GANs) [28]–[30].

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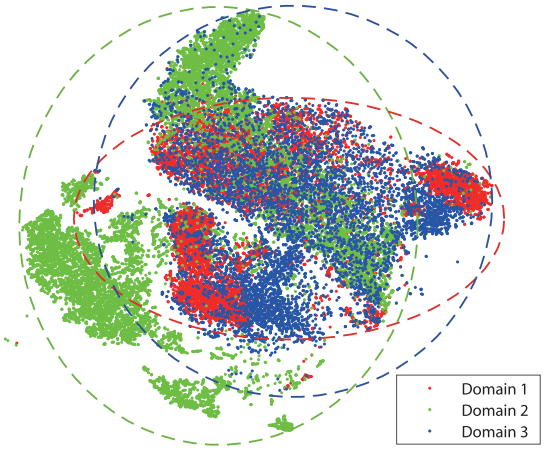


Fig. 1. Distributions of three domains are different from each other. For each domain, raw features are extracted for each pixel in one image and then concatenated with that of the corresponding pixel in the other image. We visualize the concatenated raw features by t-SNE.

Unsupervised change detection methods are widely used for flexibility in practice. However, they rely on some similarity measures or model assumptions in distinguishing between changed and unchanged classes. They also ignore the prior knowledge from existing labeled data, which makes it difficult to make a breakthrough in the accuracy of change detection maps.

Some of the aforementioned change detection methods are (semi)supervised [2], [17]. For example, Li *et al.* [31] presented a supervised change detection method with five labeled data sets to construct two coupled dictionaries. The labeled data sets were captured by different SAR sensors and contained kinds of land coverings, providing complementary information. In practice, it is difficult to collect a large amount of training data that are required in [31]. To this end, deformable dictionary learning for change detection was proposed in [32], making the coupled dictionaries generalized to a greater variety of image patterns. In our previous work [23], a transferred deep learning-based change detection framework was introduced to learn two tasks simultaneously. One was change detection in the labeled source domain, and the other was DI reconstruction in the unlabeled target domain. Fig. 1 shows the distributions of raw features extracted from different data sets (each data set is considered as one domain). It is observed that the distributions of the three domains are different from each other. The data shift across domains is derived from imaging modalities and conditions, spatial resolutions, noise and distortions, spectral and radiometric of the ground object, and so on. Because of the data shift, the performance of the change detection model trained from the source domain is degraded on the target one. To improve the change detection map of the target domain, knowledge adaptation is required to reduce the distribution discrepancy between the source and target domains.

In this article, suppose that multiple labeled scenes with change detection maps are available as source domains. Our aim is to yield a high-quality change detection map for an unlabeled new scene after knowledge adaptation from multisource domains. To this end, we propose a change detection framework based on selective adversarial adaptation. The adaptation between multisource and target domains is achieved by two domain discriminators. The first domain discriminator is designed to identify the difference between distributions of each source domain and the target one. According to the output of the first domain discriminator, we rank all the source samples. The intuition is that if one source sample is almost discriminated from the target domain, the knowledge transfer from the one is positive. Otherwise, it is negative. Thus, a DNN-based change detection model is trained with a subset of source samples that stimulate positive transfer. As for the second domain discriminator, the selected source samples are thought from one domain. Adversarial learning is introduced to further reduce the domain discrepancy between the selected source samples and the target ones. The adaption process enables the change detection model to learn a representation that is discriminative across domains. After domain adaptation, the target samples with reliable pseudolabels and the selected source ones are used to jointly fine-tune the change detection model. The reliable pseudolabels of the target domain come from a preclassification map. By making full use of the knowledge from labeled source scenes, the proposed model achieves superior change detection results on five remote sensing data sets.

The main contributions of this article are given as follows.

1) We propose a novel change detection framework based on selective adversarial adaptation. It can transfer knowledge from the existing labeled scenes to the unlabeled target one.
2) The first domain discriminator is introduced to select important samples from multisource domains to match the distribution of the target domain. It can both enhance the positive transfer and alleviate negative transfer.
3) With the second domain discriminator, the domain discrepancy between the selected source and target samples is reduced by adversarial learning.
4) For the target domain, homogeneity- and boundary-based strategies are exploited to make the pseudolabels from a preclassification map reliable. It plays an important role in the fine-tuning stage.

The rest of this article is organized as follows. Section II briefly reviews the DNN-based change detection methods and the domain adaptation approaches. Section III provides the details about the selective adversarial adaptation with two domain discriminators. Section IV describes the proposed change detection framework. In this section, we also illustrate the joint fine-tuning of the change detection model in detail. The experimental results on five remote sensing data sets as well as the discussion are provided in Section V. Section VI draws the conclusion of this article.

II. Related Work

A. Change Detection

DNN-based methods [33], [34] can learn hierarchical features from the input images in a data-driven fashion. For example, stacked Fisher autoencoder (SFAE) [20] was developed to suit the environment with the multiplicative noise

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for SAR images change detection. By embedding spatial information into the loss function of CNN, a local restricted CNN (LRCNN) was presented in [34] to enforce local similarity between neighboring pixels in the change detection maps. With siamese convolutional networks [21], the feature maps extracted from two optical aerial images were compared to get the DI, which was more separable than that generated by raw images. Mou *et al.* [35] presented a recurrent convolutional neural network (ReCNN) to learn spectral–spatial–temporal features for change detection in multispectral images. Liu *et al.* [36] proposed a symmetric convolutional coupling network (SCCN) for change detection between two heterogeneous images captured by different types of sensors. In DNN-based methods, feature learning and classifier training are integrated together. Wang *et al.* [37] presented a general end-to-end 2-D convolutional neural network (GETNET) for hyperspectral image change detection task. The overall performance was improved by end-to-end learning.

More recently, GAN and its variants are widely used in image translation tasks. In remote sensing, conditional GAN (cGAN) [28], [38] and cycle-consistent GAN (CycleGAN) [29] were proposed to translate the image from one sensor to that from another one. As a result, the image appearance difference from multisensors was mitigated, and the direct comparison was feasible. Besides, Hou *et al.* [30] proposed a change detection GAN (CDGAN), which reformulated change detection as the translation between two inputs and the change detection map.

B. Domain Adaptation

For domain adaptation, some strategies are defined to mitigate the gap between the distributions of two domains. For example, Ge and Yu [39] proposed a deep representation-based adaptation method, in which the important samples reused for the target domain are selected by searching nearest neighbors in the source domain. Ghifary *et al.* [40] designed a deep reconstruction-classification network to jointly learn a shared representation for two tasks. In [41], an appearance adaptation network was presented to learn a transformation from one domain to another one in the pixel space.

In remote sensing [42], data shift across domains exists mainly because of the difference in imaging mechanisms and characteristics of the ground objects (see Section I). DL-based domain adaptation is widely applied for cross-scene or cross-sensor land-cover classification [43], [44], target recognition [45], and scene classification [46], [47]. In [43] and [46], maximum mean discrepancy (MMD) was embedded into DNN-based methods to measure the distribution discrepancy between the source and target domains. Huang *et al.* [45] discussed what, where, and how to transfer in SAR target recognition and proposed a deep learning method based on multikernel MMD (MK-MMD) in domain adaptation. Adversarial learning [48], [49] is also popular in adapting the feature distribution from the target domain to the source domain by introducing a discriminator. Lu *et al.* [47] used multiple discriminators for multisource domain adaptation, and each discriminator was responsible for one source domain. In addition, the gradient reversal layer (GRL) [44] was inserted between feature extractor and domain discriminator to learn domain-invariant representations.

Domain adaptation has been extensively explored in multispectral [44], hyperspectral [43], [50], and aerial [46], [51] images. For example, the VGG-16 network was pretrained on ImageNet and fine-tuned by multispectral images for forest change detection [26]. Nevertheless, there are only a few studies focusing on domain adaptation to address the data shift in SAR images. Wang *et al.* [52] introduced simulated SAR data to alleviate the problem of lacking labeled training data. Due to the rapid development of deep learning, some studies have attempted to explore the knowledge transfer between optical and SAR images. Ye *et al.* [48] proposed an SAR image retrieval method based on domain-invariant features between aerial and C-band SAR images. In [53], deep features for TerraSAR image classification were extracted on the intermediate layers of the network pretrained on the CIFAR-10 data set. To the best of our knowledge, there are not yet general methods for domain adaptation between images with different types of sensors.

In this article, we try to explore knowledge transfer between optical and SAR images for change detection with DNN-based domain adaptation methods. In [23], each source sample is equally important to the target domain in the adaptation. To address this problem, we propose a domain discriminator in this article to pick out a subset of important source samples to train a DNN-based change detection model. To further reduce the distribution discrepancy between the selected source and target samples, adversarial learning is introduced between the change detection model and the second discriminator.

III. Selective Adversarial Adaptation

Goodfellow *et al*. [54] formulated the GAN as a min–max two-player game between a generator and a discriminator. The generator maps noise vectors into fake images to fool the discriminator. The discriminator distinguishes between the real images and the fake ones. The min–max game is equivalent to minimize the Jensen–Shannon divergence between the real and fake distributions. GAN and its variants [55], e.g., coupled GAN (CoGAN) [56], are widely used in image-to-image translation, image synthesis, and semantic segmentation.

In this article, two domain discriminators are applied in adapting the change detection model learned from multisource domains to the target domain (see Fig. 2). The first domain discriminator D_0 is designed to pick out a subset of important samples from the multisource domains to be reused for the target domain, which is described in Section III-A. The source samples are selected when they are almost discriminated from the target domain. The second domain discriminator D_1 is used to further minimize the discrepancy between the distributions of the selected source and target samples. The feature extractor G_f and the second domain discriminator D_1 are optimized by way of a min–max two-player game in Section III-B.

A. Sample Selection From Multisource Domains With D_0

In practice, it is likely to have multiple scenes with labels to help detect changes in the unlabeled target one.

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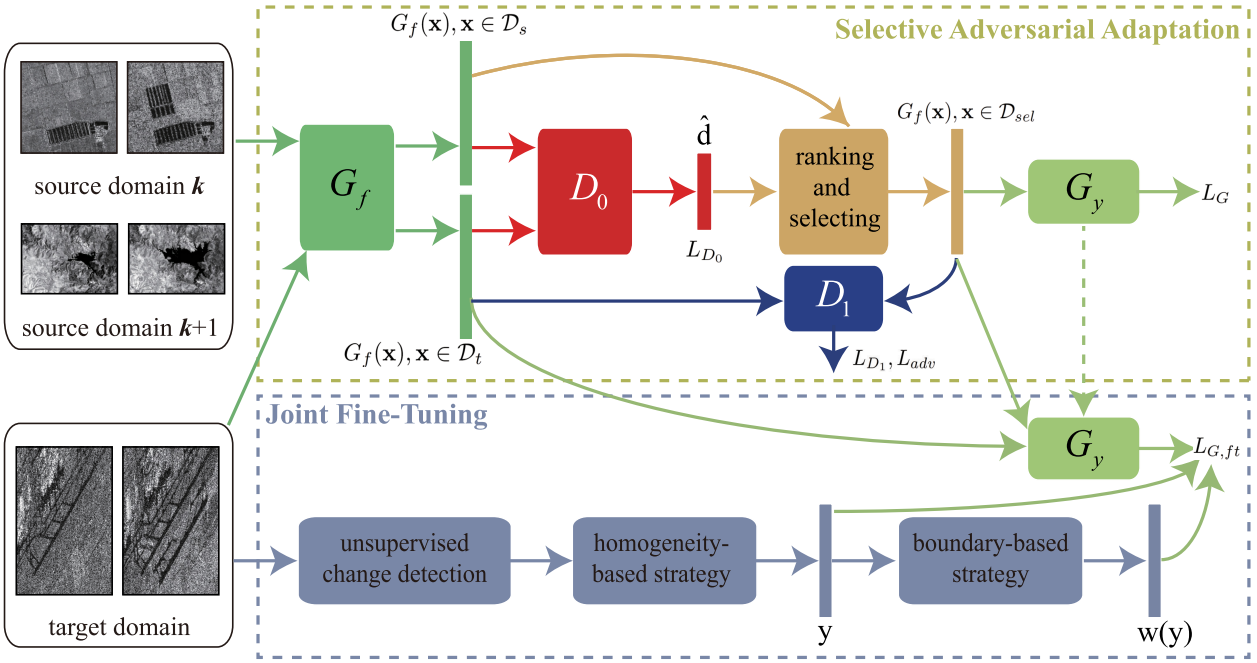


Fig. 2. Overview of the proposed change detection framework based on selective adversarial adaptation. G_f is the feature extractor for multisource and target domains. The first domain discriminator D_0 (red) built on the base of G_f can pick out important samples from multisource domains $\mathcal{D}_s = \mathcal{D}_{s_0} \cup \mathcal{D}_{s_1} \cdots \cup \mathcal{D}_{s_k} \cdots$. The domain discrepancy between the selected source samples and the target ones are further reduced by adversarial learning between the second domain discriminator D_1 (blue) and G_f. G_y is the classifier to distinguish between changed and unchanged classes. In selective adversarial adaptation, G_f is updated by ascending its stochastic gradient $(\partial L_G/\partial\theta_f) + \lambda(\partial L_{\text{adv}}/\partial\theta_f)$, and G_y is trained by the selected source samples. At the fine-tuning stage, homogeneity- and boundary-based strategies are exploited to make the pseudolabels $\mathbf{y}$ reliable. G_f and D_0 are frozen, while only G_y is fine-tuned by the selected source samples and the target ones with pseudolabels.

Multiple source domains contain much more information than one source domain. However, the distributions of multiple source data are usually different from each other (as shown in Fig. 1). It is not reasonable to learn a change detection model with all the source data available. Instance-based domain adaptation methods assume that some of the source data are useful in learning a discriminative model for the target domain, but some of them may be harmful. Therefore, it is essential to select a subset of important samples from the multisource domains to improve positive transfer.

For multisource domain adaption, the kth source domain is denoted by $\mathcal{D}_{s_k} = \{(\mathbf{x}_i^{s_k}, \mathbf{y}_i^{s_k})\}_{i=1}^{N_k}$ with N_k labeled samples. $\mathcal{D}_{s_k}$ and $\mathcal{D}_{s_{k+1}}$ are from different distributions, $P_k \neq P_{k+1}$. We are also given a target domain $\mathcal{D}_t = \{(\mathbf{x}_i^t)\}_{i=1}^{N_t}$ with N_t unlabeled samples. The distributions of the kth source domain P_k and that of the target domain Q are also different, $P_k \neq Q$. Obviously, a change detection model G learned from each of the source domains may perform badly in the target domain without adaptation.

In this article, the change detection model G includes a deep feature extractor G_f and a classifier G_y that can identify the changed pixels. With multiple labeled source domains $\mathcal{D}_{s_k}, k = 0, \ldots, n_{\text{src}} - 1$ and the unlabeled target one $\mathcal{D}_t$, G_f is designed to learn transferable features and G_y to reduce the shifts in the joint distributions across domains. Consequently, target risk is minimized, e.g., $\epsilon_t(G_f, G_y) = \Pr_{(\mathbf{x},\mathbf{y})\sim Q}[\ell(G_y(G_f(\mathbf{x})), \mathbf{y})]$. $G_y(G_f(\mathbf{x}))$ denotes the probability map for changed class with size of $m \times m$ that is produced by the change detection model given an input $\mathbf{x}$ with size of $m \times m \times 2$. n_{src} is the number of source domains.

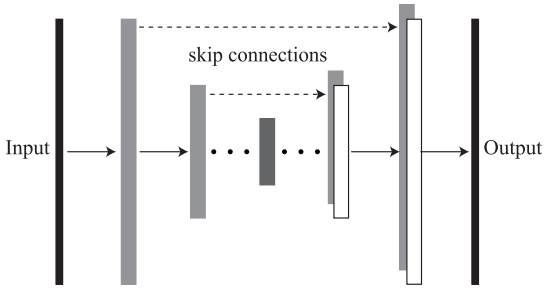


Fig. 3. U-net architecture. Skip connection exists between each layer in the contracting path and the corresponding layer in the expanding path.

The architecture of G follows the general shape of a "U-net" [57] with skip connections (as shown in Fig. 3) for its superior performance in semantic segmentation. U-net consists of a contracting path to capture spatial context and a symmetric expanding patch that enables precise localization. Each skip connection concatenates the feature maps of each layer in the contracting path and those of the corresponding layer in the expanding path.

In order to maximally match the multisource and target domains, a subset of important samples $\mathcal{D}_{\text{sel}}$ is selected from multisource domains. Then, these selected source samples are reused to train the change detection model G. Formally, we have the following loss function for G:

$$L_G(\theta_f, \theta_y) = \frac{1}{N_{\text{sel}}} \sum_{\mathbf{x}_i \in \mathcal{D}_{\text{sel}}} \ell_y(G_y(G_f(\mathbf{x}_i)), \mathbf{y}_i) \tag{1}$$

where $\mathcal{D}_{\text{sel}}$ includes N_{sel} selected source samples.

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The sample selection is guided by two principles. On one hand, the source samples that are almost discriminated from the target domain should be selected to improve the positive transfer. On the other hand, the source samples that do not match the distribution of the target samples should be discarded to alleviate negative transfer. The two principles are reminiscent of the discriminator of GANs. In this article, a domain classifier D_0 is designed to distinguish the feature distributions between each source domain and that of the target one. For D_0, each scene is regarded as an individual domain. The loss function for D_0 is defined as follows:

$$L_{D_0}(\theta_{d_0}) = \frac{1}{N_t}\sum_{\mathbf{x}_i\in\mathcal{D}_t}\ell_{d_0}(D_0(G_f(\mathbf{x}_i)),\mathbf{d}_i) + \sum_{k=0}^{n_{\text{src}}-1}\frac{1}{N_k}\sum_{\mathbf{x}_i\in\mathcal{D}_{s_k}}\ell_{d_0}(D_0(G_f(\mathbf{x}_i)),\mathbf{d}_i). \quad (2)$$

In (2), $\mathbf{d}_i$ is a vector with size of $(n_{\text{src}}+1)\times 1$. It is the one-hot encoding, indicating the domain to which $\mathbf{x}_i$ belongs. Softmax function is used in the final layer of D_0. In (1) and (2), ℓ_y and ℓ_{d_0} are binary and cross entropy losses, respectively.

According to the output of D_0, $\hat{\mathbf{d}}_i = D_0(G_f(\mathbf{x}_i))$, we rank all the training samples from multisource domains as $\{\mathbf{x}_{\text{ind}_0},\mathbf{x}_{\text{ind}_1},\ldots,\mathbf{x}_{\text{ind}_{N_s-1}}\}$ to make

$$\hat{p}_{\text{ind}_0} > \hat{p}_{\text{ind}_1} > \cdots > \hat{p}_{\text{ind}_{N_s-1}} \quad (3)$$

where $\hat{p}_{\text{ind}_0},\hat{p}_{\text{ind}_1},\ldots,\hat{p}_{\text{ind}_{N_s-1}}$ represent the probabilities that $\mathbf{x}_{\text{ind}_0},\mathbf{x}_{\text{ind}_1},\ldots,\mathbf{x}_{\text{ind}_{N_s-1}}$ are discriminated from the target domain. $N_s = N_0+N_1+\ldots+N_{n_{\text{src}}-1}$ is the number of training samples from n_{src} source domains. If one sample of the source domain is discriminated from the target domain with a high possibility, it is considered to be similar to those from the target domain. After ranking, only N_{sel} important source samples are reused as $\mathcal{D}_{\text{sel}} = \{\mathbf{x}_{\text{ind}_0},\mathbf{x}_{\text{ind}_1},\ldots,\mathbf{x}_{\text{ind}_{N_{\text{sel}}-1}}\}$. As a result, it can both enhance the positive transfer and alleviate the negative transfer since $\mathcal{D}_{\text{sel}}$ shares a similar distribution with $\mathcal{D}_t$.

In experiments, we set $N_{\text{sel}} = \alpha N_s$ and $0 < \alpha \leq 1$. If $\alpha = 0.5$, it means that only half of the source training samples are used to pretrain the change detection model G.

B. Feature Alignment With Adversarial Learning

With the first domain discriminator D_0, we select a subset of important source samples $\mathcal{D}_{\text{sel}}$. Since the samples from $\mathcal{D}_{\text{sel}}$ are almost discriminated from the target domain, they can be thought from one domain. In the next, the second domain discriminator D_1 is introduced to distinguish between $\mathcal{D}_{\text{sel}}$ and $\mathcal{D}_t$ from the pixel level.

Inspired by GANs, we use D_1 to compete with the feature extractor G_f. Adversarial learning [58] is used to learn domain-invariant features to further align the distributions of $\mathcal{D}_{\text{sel}}$ and $\mathcal{D}_t$. Since change detection is a pixel-level task, each spatial unit in the output space is considered as an individual instance. Then, the loss for D_1 is defined as follows:

$$L_{D_1}(\theta_{d_1}) = -\frac{1}{N_{\text{sel}}}\sum_{\mathbf{x}_i\in\mathcal{D}_{\text{sel}}}\sum\sum\log(D_1(G_f(\mathbf{x}_i))) - \frac{1}{N_t}\sum_{\mathbf{x}_i\in\mathcal{D}_t}\sum\sum\log(1-D_1(G_f(\mathbf{x}_i))). \quad (4)$$

There are various possible choices of adversarial loss functions. To concentrate on the target samples, we use the adversarial loss similar to GANs

$$L_{\text{adv}}(\theta_f) = -\frac{1}{N_t}\sum_{\mathbf{x}_i\in\mathcal{D}_t}\sum\sum\log(D_1(G_f(\mathbf{x}_i))). \quad (5)$$

With the adversarial loss, the loss function for the change detection model G is renewed

$$L_{G,\text{adv}}(\theta_f,\theta_y) = L_G + \lambda L_{\text{adv}} \quad (6)$$

where $\lambda > 0$ is a penalty parameter.

The target samples are not only used in minimizing $L_{G,\text{adv}}$ to train G. They are also used by two domain discriminators D_0 and D_1 to fulfill the multisource domain adaptation. The parameters $\hat{\theta}_f$, $\hat{\theta}_y$, $\hat{\theta}_{d_0}$, and $\hat{\theta}_{d_1}$ can be obtained by

$$\begin{aligned}(\hat{\theta}_f,\hat{\theta}_y) &= \arg\min_{\theta_f,\theta_y} L_{G,adv}(\theta_f,\theta_y)\\ \hat{\theta}_{d_0} &= \arg\min_{\theta_{d_0}} L_{D_0}(\theta_{d_0})\\ \hat{\theta}_{d_1} &= \arg\min_{\theta_{d_1}} L_{D_1}(\theta_{d_1}).\end{aligned} \quad (7)$$

The training processes of G, D_0, and D_1 are shown in Fig. 2 and Algorithm 1. The detailed architectures for G, D_0, and D_1 are presented in Section V-B.

IV. Proposed Change Detection Framework

Let us consider a pair of remote sensing images, $\mathbf{X}_1$ and $\mathbf{X}_2$, acquired over the same geographical area at two different times T_1 and T_2, respectively. It is assumed that $\mathbf{X}_1$ and $\mathbf{X}_2$ have been radiometrically corrected and geometrically coregistered. The size for each of the considered images is $H\times W$. Let $\Omega = \{\omega_c,\omega_u\}$ be the set of labels for change detection, where ω_c and ω_u denote the changed and unchanged classes, respectively. In this article, the ground truth $\text{GT} = \{\text{cm}(a,b), 1\leqslant a\leqslant H, 1\leqslant b\leqslant W\}$ is available for multisource domains and $\text{cm}(a,b)\in\Omega$. The labeled data from source domains can provide useful information in learning a change detection model for the target domain.

Fig. 2 shows the pipeline of the proposed change detection framework. In selective adversarial adaptation, the feature extractor G_f, the classifier G_y, and two domain discriminators D_0 and D_1 are trained. The image pairs $\mathbf{X}_1$ and $\mathbf{X}_2$ and the CD map GT from each source domain are cropped into patches with a size of $m\times m$, which are then fed into G_f. However, only a subset of source samples $\mathcal{D}_{\text{sel}}$ is reused to train G_y and D_1. These samples are selected since they are almost discriminated against from the target domain according to the output of D_0. With $\mathcal{D}_{\text{sel}}$ to train G_f and G_y, not only the positive transfer is enhanced but also the negative transfer is alleviated. Moreover, adversarial learning is introduced

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Algorithm 1 Selective Adversarial Adaptation

Input:

Labeled data $\mathcal{D}_s = \mathcal{D}_{s_0} \cup \mathcal{D}_{s_1} \cdots \cup \mathcal{D}_{s_k} \cdots$ from n_{src} source domains, where $\mathcal{D}_{s_k} = \{(\mathbf{x}_i^{s_k}, \mathbf{y}_i^{s_k})\}_{i=1}^{N_k}$; Unlabeled data from the target domain $\mathcal{D}_t = \{(\mathbf{x}_i^t)\}_{i=1}^{N_t}$; Feature Extractor G_f; The Classifier G_y; First domain discriminator D_0; Second domain discriminator D_1; Learning rate lr;

while not converged **do**
  Extract features by $G_f(\mathbf{x})$ for $\mathbf{x}$ from $\mathcal{D}_s$ and $\mathcal{D}_t$
  for each batch from $\mathcal{D}_s$ and $\mathcal{D}_t$ **do**
    Compute L_{D_0} for D_0 using Eq. (2)
    Update θ_{d_0} using Eq. (7)
  end for
  Ranking source samples from $\mathcal{D}_s$ according to Eq. (3) and selecting important samples $\mathcal{D}_{\text{sel}}$
  for each batch from $\mathcal{D}_{\text{sel}}$ and $\mathcal{D}_t$ **do**
    Compute L_{D_1} for D_1 using Eq. (4)
    Update θ_{d_1} using Eq. (7)
  end for
  for each batch from $\mathcal{D}_{\text{sel}}$ and $\mathcal{D}_t$ **do**
    Compute $L_{G,adv}$ for G using Eqs. (1), (5), (6)
    Update θ_f, θ_y using Eq. (7)
  end for
  Compute the standard deviation σ_0 and σ_1 for the average of L_{D_0} and L_{D_1} within the last 10 epoches, respectively
end while

Output:

Parameters $\hat{\theta}_f$ and $\hat{\theta}_y$ for G, $\hat{\theta}_{d_0}$ for D_0, and $\hat{\theta}_{d_1}$ for D_1; Selected source samples $\mathcal{D}_{\text{sel}}$.

between D_1 and G_f to further align the feature distributions of $\mathcal{D}_{\text{sel}}$ and $\mathcal{D}_t$. After training, the feature extractor G_f can extract discriminative features for change detection task in the target domain.

To make the most of the prior information, we also propose to fine-tune the classifier G_y by some target samples with pseudolabels. At this stage, the pseudolabels are desired to be reliable. We propose homogeneity- and boundary-based strategies for this purpose. The fine-tuning process further improves the accuracy of the change detection map for the target domain.

A. Joint Fine-Tuning

At the fine-tuning stage, the samples with pseudolabels from target domain are combined with the selected labeled source samples to fine-tune the classifier G_y of the change detection model (see Algorithm 2). At this stage, the loss function for G is defined by

$$L_{G,ft}(\theta_f, \theta_y) = \frac{1}{N_{\text{sel}}} \sum_{\mathbf{x}_i \in \mathcal{D}_{\text{sel}}} \ell_y(G_y(G_f(\mathbf{x}_i)), \mathbf{y}_i) + \frac{1}{N_t} \sum_{\mathbf{x}_i \in \mathcal{D}_t} \mathbf{w}(\mathbf{y}_i) \ell_y(G_y(G_f(\mathbf{x}_i)), \mathbf{y}_i) \quad (8)$$

where $\mathbf{w}(\mathbf{y}_i)$ is the weight map for $\mathbf{y}_i$ in the target domain. When $\mathbf{x}_i \in \mathcal{D}_t$, the corresponding $\mathbf{y}_i$ are pseudolabels from a preclassification map. Therefore, $\mathbf{w}(\mathbf{y}_i)$ is required to denote the confidence of each pixel to be used at the fine-tuning stage. In the next subsection, we will illustrate how to generate pseudolabels and weight maps for the target domain.

Algorithm 2 Fine-Tuning of the Change Detection Network

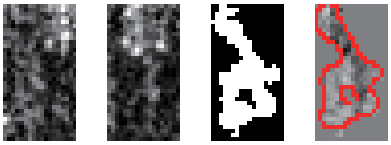
Input:

Selected labeled source data $\mathcal{D}_{\text{sel}}$; Target data $\mathcal{D}_t$ with pseudolabels; Weight maps $\mathbf{w}$; Network parameters θ_f, θ_y; Learning rate lr;

for each epoch **do**
  for each batch from $\mathcal{D}_{\text{sel}}$ and $\mathcal{D}_t$ **do**
    Compute $L_{G,ft}$ using Eq. (8)
    Update θ_y using Eq. (9)
  end for
end for

Output:

Parameters $\hat{\theta}_y$ for G.



(a) (b) (c) (d)

Fig. 4. (a)–(d) One fake changed area from the preclassification map since the corresponding difference map diff is heterogeneous. (a) and (b) Input patch pairs, patch_1 and $patch_2$. (c) Corresponding small changed areas from the preclassification map. (d) Difference map of the small changed area.

Since the features from G_f are suitable for the target domain after selective adversarial adaptation, only the parameters θ_y related to G_y are updated in the fine-tuning process. θ_y is updated by

$$\hat{\theta}_y = \arg\min_{\theta_y} L_{G,ft}(\theta_f, \theta_y). \quad (9)$$

B. Homogeneity- and Boundary-Based Strategies

In general, the target domain is fully unlabeled for change detection in remote sensing images. The preclassification maps obtained by the unsupervised methods can provide pseudolabels. However, the preclassification map is not entirely correct. The focus should be on the pixels that have a high possibility of being correctly classified. In this article, a homogeneity-based strategy is exploited to generate pseudolabels and a boundary-based strategy to compute the confidence that the pseudolabels are reliable.

1) Homogeneity: We divide the preclassification map into small changed areas based on the connectivity of each pixel [e.g., Fig. 4(c)]. Each pixel in one changed area is not connected to that in another changed area. The homogeneity is evaluated based on the difference map diff between the input patch pairs patch_1 [e.g., Fig. 4(a)] and patch_2 [e.g., Fig. 4(b)] for each small changed area bw [e.g., Fig. 4(c)]

$$\text{diff} = (\text{patch}_1 - \text{patch}_2) * \text{bw}. \quad (10)$$

For each small changed area, we assume that there is only one type of change. Then, we have the corresponding

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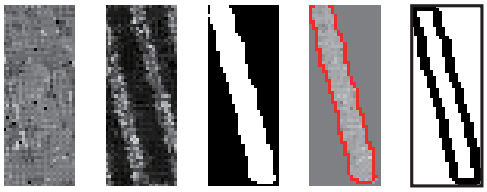


Fig. 5. (a)–(e) One homogeneous changed area. (a) and (b) Input patch pairs, patch_1 and patch_2. (c) Corresponding small changed area from the preclassification map. (d) Difference map of the small changed area. (e) Weight map for the homogeneous changed area.

difference map diff > 0 or diff < 0 for most of the pixels in each small changed area. It can be considered as the homogeneous changed area. Otherwise, we think that the small changed area from the preclassification map is resulted from large noise. In conclusion, the difference map diff of each small changed area is heterogeneous if it satisfies the following equation:

$$0.1 < \frac{\text{sum}(\text{diff} > 0)}{\text{sum}(bw)} < 0.9 \tag{11}$$

where sum(diff > 0) is the number of pixels that have positive values in diff and sum(bw) is the number of the changed pixels in bw. For heterogeneous changed area, we update it into the unchanged class. One fake changed area from the preclassification map is shown in Fig. 4(c) as the corresponding map diff in Fig. 4(d) is heterogeneous. Fig. 5(c) shows the homogeneous changed area since the corresponding map diff > 0 for most of the pixels [shown in (d)].

2) Boundary: In the preclassification map, we observe that the pixels close to the center of the changed or unchanged areas always have a high possibility to be correctly classified. On the contrary, the pixels close the border between the changed and unchanged areas are difficult to be correctly classified. Based on this observation, the loss function in (8) for the target domain at the fine-tuning stage is regularized by a weight map. The pixels close to the center of the changed or unchanged areas have the confidence to fine-tune the change detection network, while the pixels close to the border between the changed and unchanged areas do not have the confidence.

The contextual information of each pixel is used to decide whether it is located close to the center of the changed or unchanged areas. For the pixel at position (a, b), let $\mathcal{N}_{ab}$ be the neighborhood with size of $r \times r$. Then, a sum of the changed pixels in $\mathcal{N}_{ab}$ is computed, sum_{ab}. When $\text{sum}_{ab} \leqslant \delta$, the pixel (a, b) is considered to be close to the center of the unchanged area. $0 < \delta < r^2$ is a constant value. When $\text{sum}_{ab} \geqslant r^2 - \delta$, it means the pixel (a, b) is close to the center of the changed area. If $\delta < \text{sum}_{ab} < r^2 - \delta$, the pixel (a, b) is located on the border between the changed and unchanged areas. To sum up, a weight map $\mathbf{w} = \{w(a, b), 1 \leqslant a \leqslant m, 1 \leqslant b \leqslant m\}$ is precomputed by

$$w(a, b) = \begin{cases} 1, & \text{sum}_{ab} \geqslant r^2 - \delta \text{ or } \text{sum}_{ab} \leqslant \delta \\ 0, & \text{else.} \end{cases} \tag{12}$$

In this article, both r and δ are set to 5 empirically. Fig. 5(i) shows the weight map for the changed area of Fig. 5(g).

C. Inference for the Testing Images

At the testing stage, the image pairs $\mathbf{X}_1$ and $\mathbf{X}_2$ from the target domain are also cropped into overlapping patches. Then, the CD map for each patch pairs is obtained by $\hat{\mathbf{y}}_i = G_y(G_f(\mathbf{x}_i, \hat{\theta}_f), \hat{\theta}_y)$. To get the entire CD map between $\mathbf{X}_1$ and $\mathbf{X}_2$, the overlapping area between two adjacent patches is averaged to avoid the block artifacts.

V. Experimental Results and Discussion

In this section, the proposed change detection method is evaluated on five remote sensing data sets. In order to validate the efficacy of the two domain discriminators, the comparative experiments include the following.

1) *Change Detection Network for Source Domain (G_{SD}):* It trains G only using the source data, and then, the trained G is directly applied to the target domain. It is considered as our baseline.
2) *Change Detection Network With the First Domain Discriminator (G_{D_0}):* It trains G only using a subset of source samples that are selected according to the output of D_0.
3) *Change Detection Network With Two Domain Discriminators ($G_{D_{0,1}}$):* It trains G, D_0, and D_1 simultaneously. The domain discrepancy between the target and selected source samples is reduced with adversarial learning.
4) *Fine-Tuning (FT-$G_{D_{0,1}}$):* After selective adversarial adaptation with D_0 and D_1, G is further fine-tuned by the reliable labels of the target domain.

A. Data Set Descriptions

The **Yellow River** data set includes two SAR images acquired at the region of the Yellow River Estuary in China. They were captured by Radarsat-2 in June 2008 and June 2009, respectively. It is noted that the latter SAR image contains larger speckle-noise than the former. The speckle noise will complicate change detection by increasing false alarms and missed detections. The size of two original SAR images is 7666×7692 [as shown in Fig. 6(a) and (b)]. In order to validate the proposed and comparison methods, we select two spatially disjointed subsets. We denote the two selected areas as River with a size of 291×444 and Farmland with a size of 306×291, respectively. Two temporal SAR images and the reference for River data are shown in Fig. 6(c)–(e) and those for Farmland data in Fig. 6(f)–(h).

The **Sardinia** data set was acquired at Sardinia in Italy (as shown in Fig. 7). The two TM images with a size of 412×300 were derived from the near-infrared band by Landsat-5. They provide us a view of Lake Mulargia in September 1995 and July 1996, respectively.

The **De Gaulle airport** data set shown in Fig. 8 was acquired at de Gaulle airport in France by ERS-1. Two SAR images were taken in July 1997 and October 1998, respectively. The size of each SAR image is 240×370.

The **Ottawa** data set includes two SAR images acquired by Radarsat-1 in May 1997 and August 1997, respectively. The size is 290×350. As shown in Fig. 9, they present the area over the city of Ottawa flooded.

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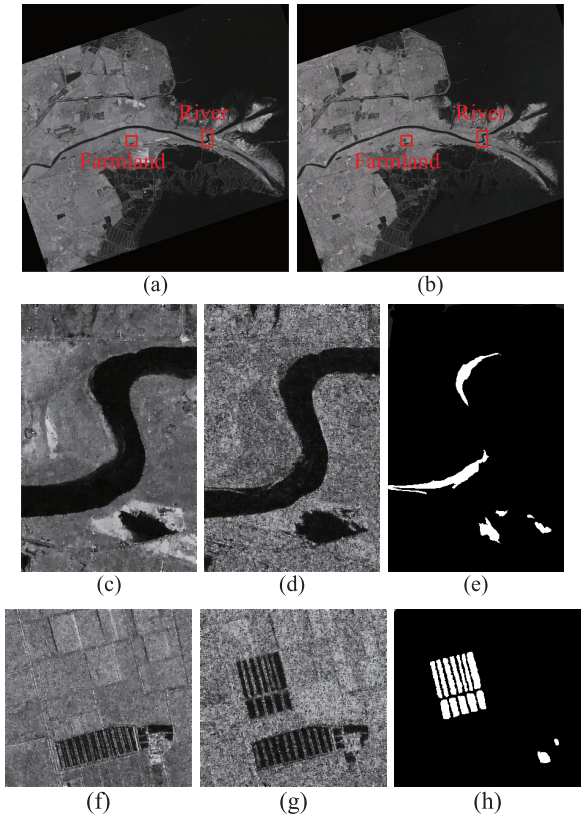


Fig. 6. SAR images relating to Yellow River Estuary. (a) Image acquired in June 2008. (b) Image acquired in June 2009. (c) River data acquired in June 2008. (d) River data acquired in June 2009. (e) Reference image for River data. (f) Farmland data acquired in June 2008. (g) Farmland data acquired in June 2009. (h) Reference image for the Farmland data.

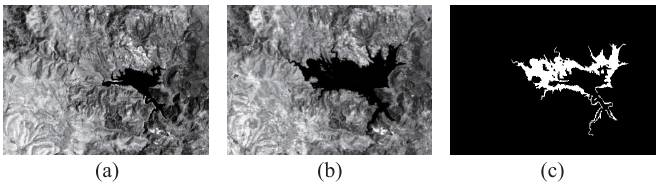


Fig. 7. Sardinia data set. (a) Image acquired in September 1995. (b) Image acquired in July 1996. (c) Reference image for the Sardinia data.

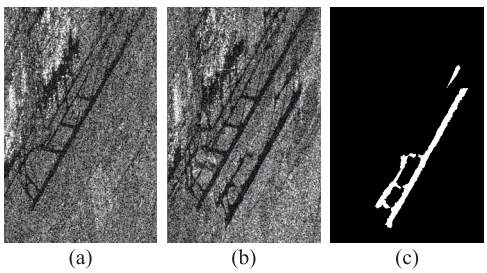


Fig. 8. De Gaulle airport data set. (a) Image acquired in July 1997. (b) Image acquired in October 1998. (c) Reference image for the de Gaulle airport data set.

The **Mexico** data set in Fig. 10 was derived from band 4 of Landsat-7 ETM+. The two temporal images with a size of 512×512 were taken at urban Mexico in April 2000 and May 2002, respectively. The change was caused by the variation of vegetation after a forest fire.

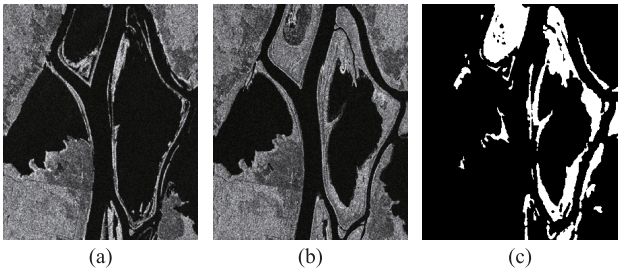


Fig. 9. Ottawa data set. (a) Image acquired in May 1997. (b) Image acquired in August 1997. (c) Reference image for the Ottawa data set.

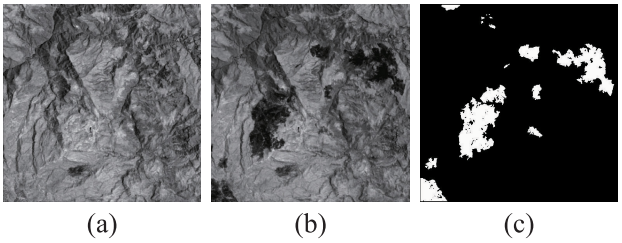


Fig. 10. Mexico data set. (a) Image acquired in April 2000. (b) Image acquired in May 2002. (c) Reference image for the Mexico data set.

The Yellow River, de Gaulle airport, and Ottawa data sets are captured by SAR sensors, while the Sardinia and Mexico data sets by optical ones. For all the data sets, the reference image was created by integrating prior information with photo interpretation based on the input images.

The descriptions for all data sets are provided in Table I. To evaluate the proposed method, imaging conditions (e.g., types of sensors and spatial resolutions) of the target domain are not identical to those of source domains.

B. Experimental Setup

1) Evaluation Measures: We follow five common measures to evaluate the predicted change detection maps, including FP, FN, OE, PC, and KC [59]. False positives (FP) error is the number of unchanged pixels that are detected as changed. False negatives (FN) error represents the number of changed pixels that are detected as unchanged. Overall error (OE) is the sum of FP and FN. The percentage of correct classified (PC) pixels is given by $\mathrm{PC} = 1 - (\mathrm{FP} + \mathrm{FN})/N$, where N is the number of all the pixels to be classified. Last but not least, Kappa coefficient (KC) is generally thought to be a more robust measure that takes into account the possibility of the agreement occurring by chance. KC is calculated by $\mathrm{KC} = (\mathrm{PC} - \mathrm{PE})/(1 - \mathrm{PE})$, where $\mathrm{PE} = (\mathrm{TP} + \mathrm{FP})(\mathrm{TP} + \mathrm{FN})/N^2 + (FN + TN)(TN + FP)/N^2$.

2) Network Architecture and Training: Referring to Fig. 2, there are three subnetworks to be designed, including the change detection network and two domain discriminators. To enable dense predictions, deconvolutional layers are included in the change detection network and the second domain discriminator. For both convolutional and deconvolutional layers, the size of the kernel is 3×3. For max-pooling

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TABLE I
DATA SETS DESCRIPTIONS

Attributes / Data Sets	Satellite	Sensor Type	Spatial Resolution	Band
River/Farmland	Radarsat-2	SAR	8m	/
Sardinia	Landsat-5	Optical	30m	4 (TM)
de Gaulle airport	ERS-1	SAR	30m	/
Ottawa	Radarsat-1	SAR	10m	/
Mexico	Landsat-7	Optical	30m	4 (ETM+)

TABLE II
SIX DOMAIN ADAPTATION TASKS FOR CHANGE DETECTION

Source / Target	Source 1	Source 2	Labeled Source Samples for Training	Unlabeled Target Samples for Training	Target Samples for Testing
River	Sardinia	de Gaulle airport	46,922	17,692	9,120
Farmland	Sardinia	de Gaulle airport	46,922	12,974	6,080
Sardinia	Farmland	de Gaulle airport	48,482	11,414	8,769
de Gaulle airport	Farmland	Sardinia	24,388	35,508	6,138
Ottawa	River	Mexico	39,218	21,926	7,068
Mexico	Farmland	Ottawa	34,900	21,526	19,600

layers, 2×2 is used. The strides for convolutional, max-pooling, and deconvolutional layers are set to 1, 2, and 2, respectively.

3) Change Detection Network G: Building upon a good baseline model is of crucial importance to achieve high-quality change detection maps. U-net architecture [57] is adopted as our baseline network. Different from [57], our change detection network has fewer layers since the size of our input patch is relatively small. The change detection network includes the following layers: C(8)-P-C(16)-P-C(32)-D(16)-C(16)-D(8)-C(1).

4) First Domain Discriminator D_0: The detailed architecture of D_0 is C(16)-P-C(32)-P-FC(128)-FC($n_{\text{src}} + 1$). For the last layer, the softmax function is used. The output of D_0 is a vector with size of $(n_{\text{src}} + 1) \times 1$. In the abovementioned shorthand notations, C(n) denotes a convolutional layer with n filters, P is a max-pooling layer, D(n) denotes a deconvolutional layer with n filters, and FC(n) is the fully connected layer with n nodes.

5) Second Domain Discriminator D_1: D_1 is comprised of two convolutional layers and two deconvolutional layers with 16, 16, 8, and 1 channels, respectively.

In this article, all the subnetworks are trained from scratch with TensorFlow. The training process for G, D_0, and D_1 is shown in Algorithm 1. With D_0, only important source samples are selected, which are then combined with the target ones to update G and D_1. Different from D_1, D_0 is trained with all source and target samples with domain labels as supervision. In the fine-tuning process (see Algorithm 2), the feature extractor G_f and the first domain discriminator D_0 are frozen. Nevertheless, only the classifier G_y is updated with the selected labeled source samples and the target ones with reliable pseudolabels. The number of training and testing samples is shown in Table II for each task. In the source domain, the patch pairs are used as training samples if they contain some changed pixels or unchanged ones that are easy to be classified into the changed class incorrectly. In the target domain, the patch pairs with a large difference between them are used to train the networks since they have rich information of changed and unchanged classes. The batch size is 64 for both the source and target domains. We initialize the convolutional and deconvolutional weights with the Xavier initializer, bias with constant value 0, and use a learning rate of 0.01. The threshold for σ_0 and σ_1 is set to 0.008 and 0.002, respectively.

6) Parameters: A systematic search is done to investigate the effects of patch size m on the change detection network. In this experiment, the River data set is divided into two parts. Then, one part is used for training and the other for testing, and vice versa. In this case, the training data and testing data come from the same domain. Otherwise, the domain discrepancy can also affect the performance of the change detection network. Fig. 11 shows the accuracy of the change detection map when m is set in the range $\{16, 28, 48, 64\}$. It can be seen that the performance of the change detection network varies little when m is selected from $\{16, 28, 48\}$. However, when m is set to 64, the change detection network performs worst. In the following experiments, patch size m is fixed to 28.

In the proposed method, α is a parameter that indicates the number of the selected source samples $N_{\text{sel}} = \alpha N_s$. Fig. 12 shows the performance of the G_{D_0} on all data sets when α is selected from $\{0.1, 0.3, 0.5, 0.7, 0.9\}$. The source domains for each target data set are given in Table II. We observe that the change detection maps are improved more when α is set between 0.3 and 0.7. When α is too large, $\mathcal{D}_{\text{sel}}$ contains a number of source samples that enable the negative transfer. When α is too small, $\mathcal{D}_{\text{sel}}$ contains only a small set of source samples that stimulate positive transfer, while a suitable α

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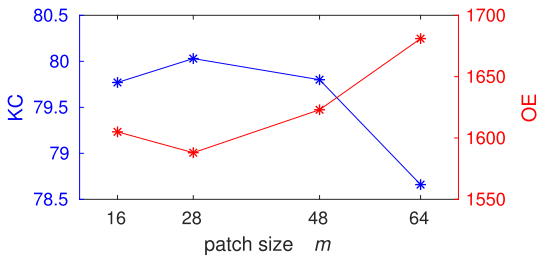


Fig. 11. Kappa and OE vary with a different patch size m.

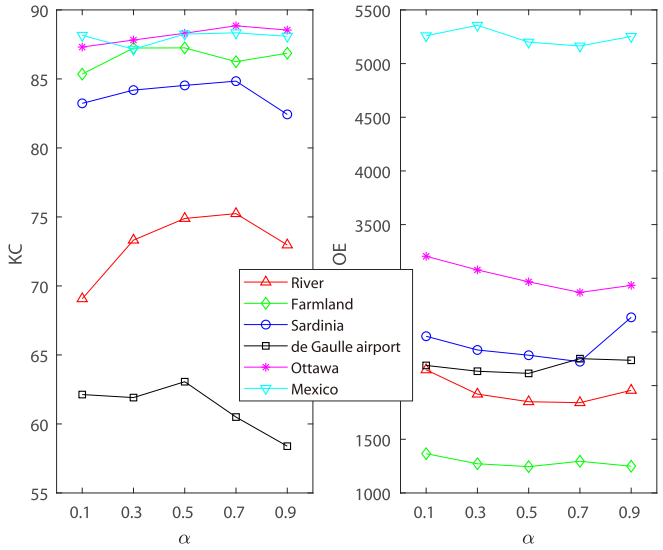


Fig. 12. Performance of G_{D_0} varies with different α's on the River, Farmland, Sardinia, de Gaulle airport, Ottawa, and Mexico data sets.

TABLE III
CHANGE DETECTION RESULTS ON THE RIVER DATA SET BY G_{D_1} USING DIFFERENT VALUES OF λ

Method		$KC(\%)$	$PC(\%)$	FP	FN	OE
G_{SD}		70.94	98.36	544	1569	2113
G_{D_1}	λ = 0.002	69.41	98.17	925	1444	2369
	λ = 0.02	71.83	98.39	585	1494	2079
	λ = **0.2**	**75.74**	**98.48**	**869**	**1080**	**1949**
	λ = 1.0	73.75	98.38	922	1176	2098
	λ = 10.0	72.27	98.41	599	1460	2059

improves the positive transfer as well as alleviates negative transfer. For convenience, α is set to 0.5 in this article.

In order to validate the efficacy of the second domain discriminator D_1, a comparative study is made between G_{SD} and G_{D_1} on two tasks: de Gaulle airport → River and Farmland → de Gaulle airport. G_{D_1} trains G and D_1 simultaneously. In this case, G is trained with all the source samples from one domain and constrained by adversarial loss. Because of the data shift between two source domains, only one source domain is used in G_{D_1}. To investigate the effects of the parameter λ, the performance of G_{D_1} is shown in Tables III and IV as $\lambda \in \{0.002, 0.02, 0.2, 1.0, 10.0\}$. From Tables III and IV, it can be seen that the change detection maps

TABLE IV
CHANGE DETECTION RESULTS ON THE DE GAULLE AIRPORT DATA SET BY G_{D_1} USING DIFFERENT VALUES OF λ

Method		$KC(\%)$	$PC(\%)$	FP	FN	OE
G_{SD}		55.49	96.92	1241	1491	2732
G_{D_1}	λ = 0.002	55.67	97.07	1024	1574	2598
	λ = 0.02	58.55	97.28	916	1500	2416
	λ = 0.2	60.98	97.20	1236	1246	2482
	λ = **1.0**	**60.07**	**97.42**	**804**	**1487**	**2291**
	λ = 10.0	60.47	97.12	1336	1223	2559

TABLE V
CHANGE DETECTION RESULTS ON THE RIVER DATA

Method	$KC(\%)$	$PC(\%)$	FP	FN	OE
GKI	68.74	97.55	2595	569	3164
RFLICM	76.01	98.32	1525	646	2171
PCANet	75.23	98.52	683	1229	1912
D_RFLICM	77.24	98.47	1227	748	1975
RDN	**78.71**	**98.77**	**378**	**1209**	**1587**
ED-DMM-UDA	75.04	98.55	548	1325	1873
G_{SD}	67.47	98.30	298	1894	2192
G_{D_0}	74.89	98.57	464	1386	1850
$G_{D_{0,1}}$	76.00	98.63	436	1337	1773
FT-$G_{D_{0,1}}$	**80.07**	**98.75**	**753**	**866**	**1619**

TABLE VI
CHANGE DETECTION RESULTS ON THE FARMLAND DATA

Method	$KC(\%)$	$PC(\%)$	FP	FN	OE
GKI	80.05	97.81	1017	931	1948
RFLICM	85.26	98.37	692	761	1453
PCANet	85.53	98.56	52	1232	1284
D_RFLICM	**89.50**	**98.86**	**352**	**661**	**1013**
RDN	88.42	98.74	430	692	1122
ED-DMM-UDA	87.55	98.67	350	832	1182
G_{SD}	85.44	98.48	332	1023	1355
G_{D_0}	87.25	98.60	535	710	1245
$G_{D_{0,1}}$	87.40	98.65	371	828	1199
FT-$G_{D_{0,1}}$	89.08	98.86	168	849	1017

are improved with different values of λ. The experimental results demonstrate that the domain discrepancy is reduced with the adversarial learning between G_f and D_1. Based on the experimental results shown in Tables III and IV, λ is set to 0.2 for $G_{D_{0,1}}$ when the River, Farmland, Sardinia, and Mexico data sets are used as target domains. For the de Gaulle airport and Ottawa data sets as target domains, λ is set to 1.0 for $G_{D_{0,1}}$.

C. Multisource Domain Adaptation With Two Domain Discriminators

For each target data set, the corresponding source domains are listed in Table II. Note that the Farmland data set is not used as the source domain for River data set, and vice versa, since they are cropped from the same scene. The proposed method is validated on six tasks listed in Table II.

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TABLE VII
CHANGE DETECTION RESULTS ON THE SARDINIA DATA

Method	$KC(\%)$	$PC(\%)$	FP	FN	OE
GKI	81.48	97.68	2088	775	2863
RFLICM	83.41	98.06	1269	1126	2395
PCANet	82.59	97.80	2124	600	2724
D_RFLICM	84.88	98.29	883	1234	2117
RDN	85.36	98.33	924	1143	2067
ED-DMM-UDA	84.27	98.11	1480	857	2337
G_{SD}	81.35	97.77	1664	1097	2761
G_{D_0}	84.53	98.15	1394	889	2283
$G_{D_{0,1}}$	85.62	98.31	1157	929	2086
FT-$G_{D_{0,1}}$	**85.94**	**98.36**	**1094**	**937**	**2031**

TABLE VIII
CHANGE DETECTION RESULTS ON THE DE GAULLE AIRPORT DATA

Method	$KC(\%)$	$PC(\%)$	FP	FN	OE
GKI	40.43	93.04	5259	925	6184
RFLICM	46.04	94.07	4460	802	5262
PCANet	25.20	84.54	13338	389	13727
D_RFLICM	57.16	97.47	521	1729	2250
RDN	62.76	97.61	704	1421	2125
ED-DMM-UDA	63.71	97.81	425	1519	1944
G_{SD}	56.92	97.35	693	1664	2357
G_{D_0}	63.06	97.62	708	1406	2114
$G_{D_{0,1}}$	64.50	97.79	532	1433	1965
FT-$G_{D_{0,1}}$	**67.49**	**97.86**	**666**	**1234**	**1900**

TABLE IX
CHANGE DETECTION RESULTS ON THE OTTAWA DATA

Method	$KC(\%)$	$PC(\%)$	FP	FN	OE
GKI	88.39	96.94	1338	1765	3103
RFLICM	89.36	97.34	151	2549	2700
PCANet	**93.26**	**98.22**	**745**	**1062**	**1807**
D_RFLICM	90.87	97.62	772	1640	2412
RDN	90.21	97.45	899	1694	2593
ED-DMM-UDA	90.27	97.47	785	1778	2563
G_{SD}	87.11	96.54	1941	1575	3516
G_{D_0}	88.31	97.08	289	2678	2967
$G_{D_{0,1}}$	89.64	97.36	466	2209	2675
FT-$G_{D_{0,1}}$	90.61	97.60	381	2050	2431

TABLE X
CHANGE DETECTION RESULTS ON THE MEXICO DATA

Method	$KC(\%)$	$PC(\%)$	FP	FN	OE
GKI	86.60	97.40	6380	447	6827
RFLICM	86.68	97.84	575	5093	5668
PCANet	86.52	97.80	743	5020	5763
D_RFLICM	89.23	98.18	1140	3621	4761
RDN	89.41	98.20	1411	3318	4729
ED-DMM-UDA	88.40	98.03	1608	3567	5175
G_{SD}	87.09	97.86	1125	4489	5614
G_{D_0}	88.25	98.02	1413	3787	5200
$G_{D_{0,1}}$	89.27	98.18	1388	3395	4783
FT-$G_{D_{0,1}}$	**89.82**	**98.29**	**976**	**3517**	**4493**

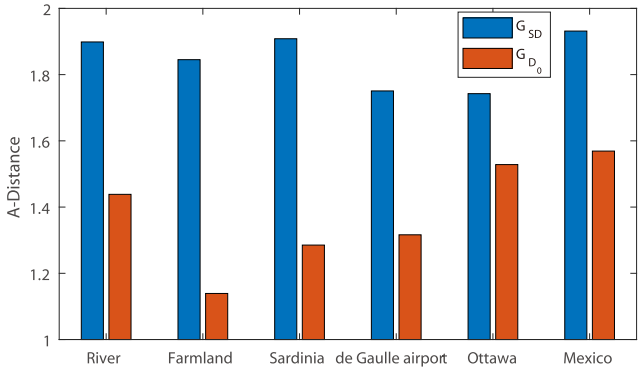


Fig. 13. $\mathcal{A}$-distance on features generated by G_{SD} and G_{D_0}, respectively.

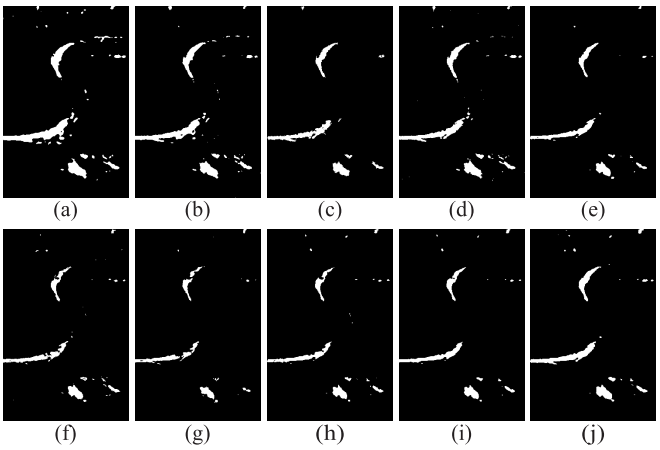


Fig. 14. CD maps for the River data set by (a) GKI, (b) RFLICM, (c) PCANet, (d) D_RFLICM, (e) RDN, (f) ED-DMM-UDA, (g) G_{SD}, (h) G_{D_0}, (i) $G_{D_{0,1}}$, and (j) FT-$G_{D_{0,1}}$.

The quantitative evaluations of the change detection maps are summarized for the proposed and comparison methods. They are provided in Tables V–X for the River, Farmland, Sardinia, de Gaulle airport, Ottawa, and Mexico data sets, respectively. We first examine the effectiveness of the first domain discriminator D_0. Compared with G_{SD}, G_{D_0} improves the change detection maps by only using the important source samples to train G. The improvement to KC ranges from 1.2% to 7.4% on all the data sets. On the de Gaulle airport data set, there is a substantial decrease in missed detections by G_{D_0} (1664) compared with G_{SD} (1406). However, false alarms are increased slightly (from 693 to 708). The corresponding change detection maps are shown in Figs. 14–19. From Figs. 14(h)–19(h), it can be seen that the number of misclassified pixels is reduced. Especially, on the Sardinia data set, the FP and FN are both reduced significantly.

$\mathcal{A}$-Distance is a measure of domain discrepancy according to domain adaptation theory in [60]. The proxy $\mathcal{A}$-distance is defined as $\hat{d}_{\mathcal{A}} = 2(1 - 2\epsilon)$, where ϵ is the generalization error of a classifier (e.g., kernel SVM) trained to distinguish

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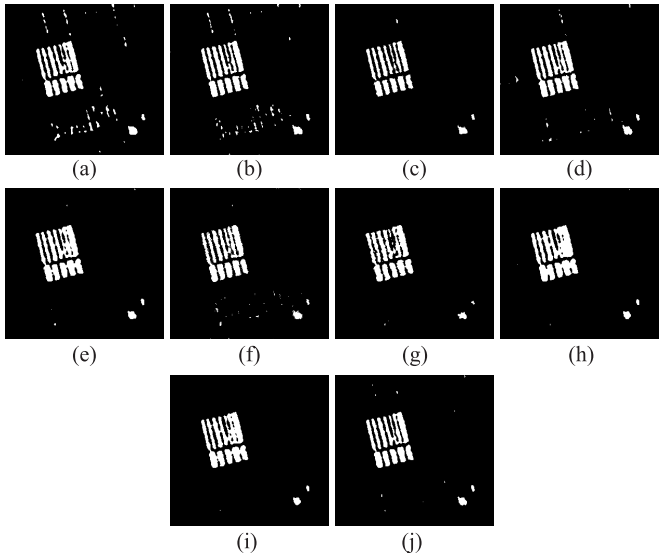


Fig. 15. CD maps for the Farmland data set by (a) GKI, (b) RFLICM, (c) PCANet, (d) D_RFLICM, (e) RDN, (f) ED-DMM-UDA, (g) G_{SD}, (h) G_{D_0}, (i) $G_{D_{0,1}}$, and (j) FT-$G_{D_{0,1}}$.

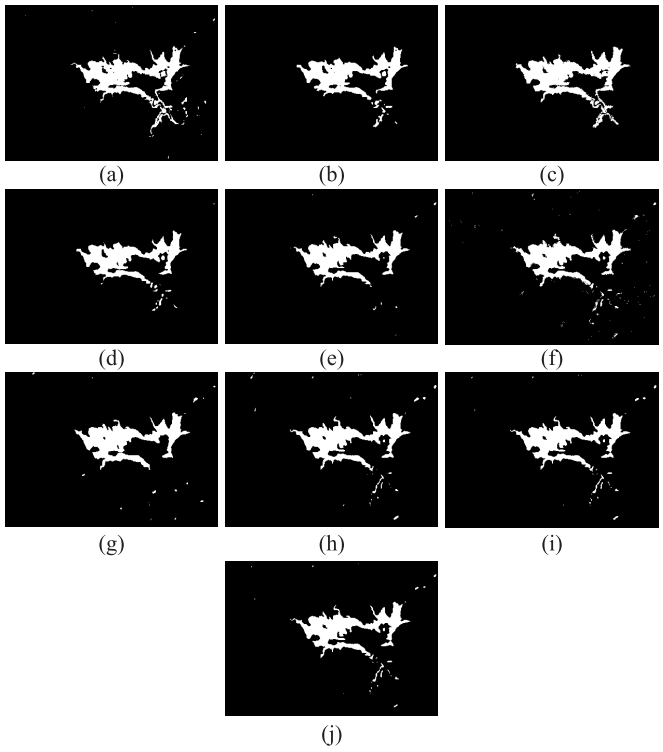


Fig. 16. CD maps for the Sardinia data set by (a) GKI, (b) RFLICM, (c) PCANet, (d) D_RFLICM, (e) RDN, (f) DMM-UDA, (g) G_{SD}, (h) G_{D_0}, (i) $G_{D_{0,1}}$, and (j) FT-$G_{D_{0,1}}$.

one domain from others. The proxy $\mathcal{A}$-distance is computed on features $G_f(\mathbf{x})$ generated by G_{SD} and G_{D_0}, respectively. From Fig. 13, the $\mathcal{A}$-distance on $G_f(\mathbf{x})$ generated by G_{SD} is smaller than that by G_{D_0}. It indicates that the domain discrepancy is reduced when only a subset of important multisource samples is selected to train the change detection network.

The second domain discriminator D_1 is used in reducing the domain discrepancy between the selected source samples

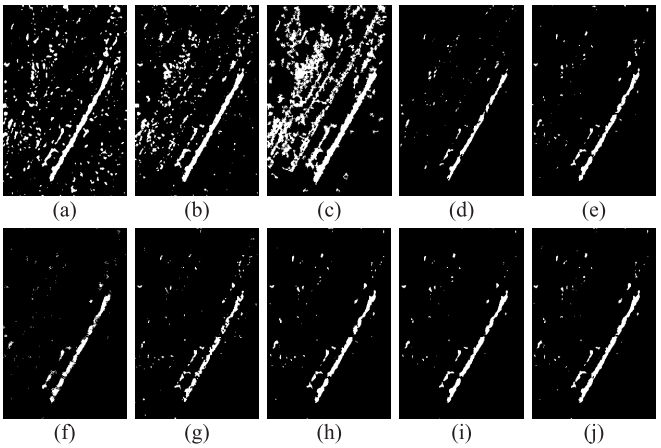


Fig. 17. CD maps for the de Gaulle airport data set by (a) GKI, (b) RFLICM, (c) PCANet, (d) D_RFLICM, (e) RDN, (f) ED-DMM-UDA, (g) G_{SD}, (h) G_{D_0}, (i) $G_{D_{0,1}}$, and (j) FT-$G_{D_{0,1}}$.

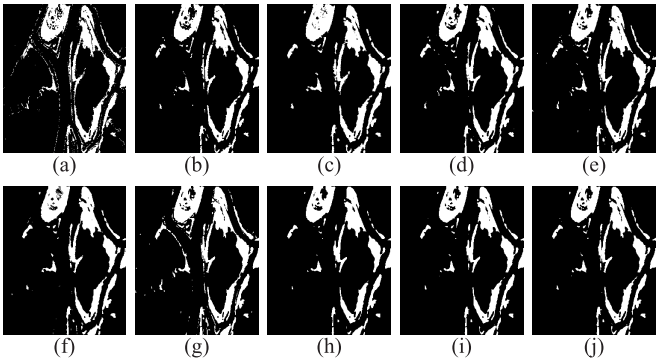


Fig. 18. CD maps for the Ottawa data set by (a) GKI, (b) RFLICM, (c) PCANet, (d) D_RFLICM, (e) RDN, (f) ED-DMM-UDA, (g) G_{SD}, (h) G_{D_0}, (i) $G_{D_{0,1}}$, and (j) FT-$G_{D_{0,1}}$.

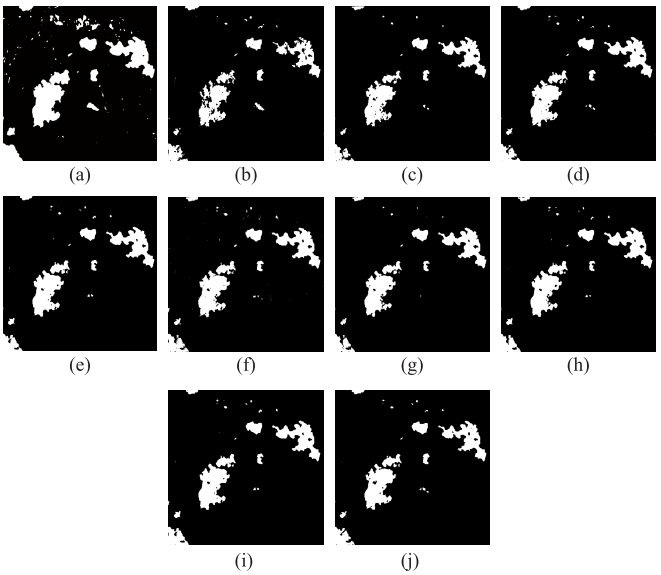


Fig. 19. CD maps for the Mexico data set by (a) GKI, (b) RFLICM, (c) PCANet, (d) D_RFLICM, (e) RDN, (f) ED-DMM-UDA, (g) G_{SD}, (h) G_{D_0}, (i) $G_{D_{0,1}}$, and (j) FT-$G_{D_{0,1}}$.

and the target samples with adversarial learning. The efficacy of D_1 has been demonstrated in Tables III and IV between G_{SD} and G_{D_1}. The qualitative and quantitative change

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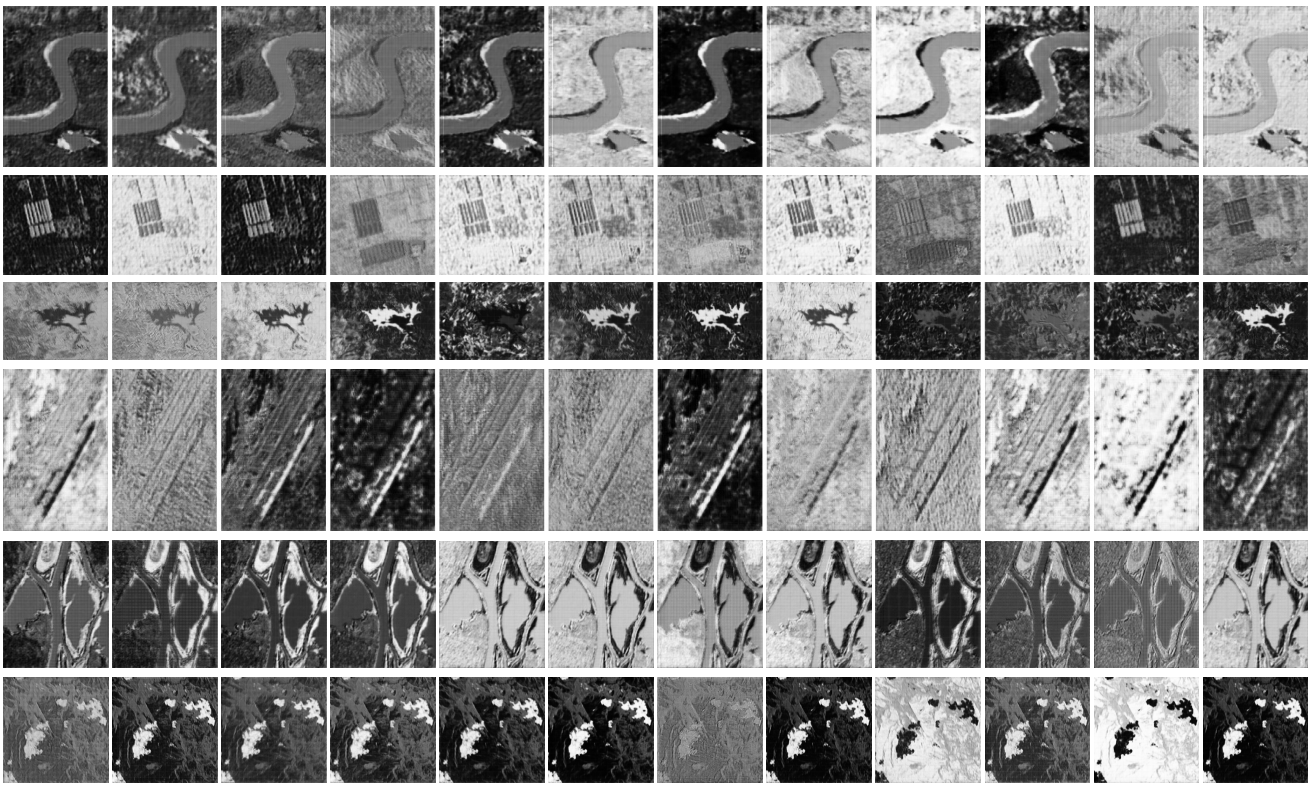


Fig. 20. Feature maps extracted from the last but one convolutional layer of the change detection network. From top to bottom: River, Farmland, Sardinia, de Gaulle airport, Ottawa, and Mexico data sets.

detection results of $G_{D_{0,1}}$ are shown in Figs. 14(i)–19(i) and Tables V–X, respectively. For the River and Sardinia data sets, $G_{D_{0,1}}$ increases the KC by about 1.1% compared with G_{D_0}. It is obvious that the domain discrepancy between the selected source samples and the target samples is smaller than that between all the source samples and the target samples. Therefore, D_1 shows a relative small effect on G_{D_0}.

D. Fine-Tuning

The change detection network after multisource domain adaptation can be fine-tuned by the reliable pseudolabels from the preclassification map of the target domain. We denote it as FT-$G_{D_{0,1}}$. To be fair, RFLICM is used as the unsupervised change detection method to obtain the preclassification map since D_RFLICM acts as one of the comparison methods. The change detection results by FT-$G_{D_{0,1}}$ are shown in Figs. 14(j)–19(j) and Tables V–X. With reliable labels from the target domain, FT-$G_{D_{0,1}}$ further improves the change detection maps. Taking River data set as an example, the KC is improved from 76.00% to 80.07% compared with $G_{D_{0,1}}$. For the Farmland data set, FT-$G_{D_{0,1}}$ removes some false alarms (from 371 to 168). It demonstrates that the proposed homogeneity- and boundary-based strategies are effective in selecting the pixels that have a large possibility of being correctly classified from the preclassification map.

E. Comparison With State-of-the-Art Methods

The proposed method is also compared with the state-of-the-art change detection methods, GKI [9], PCANet [61], RFLICM [13], and DNN-based methods [19], [23], [62]. GKI and RFLICM are implemented using the default parameters that are provided in [9] and [13], respectively. For PCANet, we use the publicly available code of [61]. We also compare the proposed method with D_RFLICM [19]. D_RFLICM is a DNN-based change detection method. In D_RFLICM, the size of the neighborhood is set to 5 as in [19]. RDN [23] proposed a reconstruction-detection network to adapt the concept of change from labeled source domains to the target one. ED-DMM-UDA [62] presented a deep metric model to learn embedding features for hyperspectral image classification. It can be learned from a source scene and transferred to the target one by minimizing adversarial loss and center magnet loss. When ED-DMM-UDA is applied in change detection, the patch size is set to 5 as in [62] (except the de Gaulle airport data set in which it is set to 7).

In the existing algorithms, GKI shows poor results with large FP. PCANet performs badly on the de Gaulle airport data set. In PCANet, a coarse-to-fine FCM algorithm is used to obtain a preclassification CD map. A large number of unchanged pixels are misclassified into the changed class in the preclassification map of the de Gaulle airport data set. However, these false alarms are not removed in the final CD maps. On all the data sets, the performance of D_RFLICM with deep representations is superior to that of RFLICM. FT-$G_{D_{0,1}}$ makes use of the knowledge from both the source and target domains. Therefore, it outperforms the state-of-the-art unsupervised change detection methods on the River, Sardinia, de Gaulle airport, and Mexico data sets.

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On the Farmland and Ottawa data sets, the performance of FT-$G_{D_{0,1}}$ is a little worse than that of D_RFLICM.

The proposed method is also compared with two deep learning methods based on domain adaptation, namely, RDN and ED-DMM-UDA. From Tables VI–X, it can be seen that FT-$G_{D_{0,1}}$ achieves better performance than RDN on the Farmland, Sardinia, de Gaulle airport, Ottawa, and Mexico data sets. The proposed method can pick out a subset of important source samples to train a change detection model, while RDN treats each source sample equally. On all data sets, FT-$G_{D_{0,1}}$ yields better change detection results than ED-DMM-UDA since the proposed method utilizes the reliable pseudolabels from a preclassification map of the target domain to fine-tune the change detection model.

In Fig. 20, we visualize the feature maps extracted from the last but one convolutional layer of the change detection network for all the data sets. From Fig. 20, it can be seen that the change detection network can learn high-level features that highlight the changed areas.

VI. Conclusion

This article presents a new change detection framework in remote sensing images, based on selective adversarial adaptation. Our main contribution is to make use of the knowledge from multiple source domains to help the change detection in the target domain. We introduce the first domain discriminator to select a subset of important source samples and adversarial learning to align the target and selected source samples with the second domain discriminator. Using the important source samples to train the change detection network, not only the positive transfer is enhanced but also the negative transfer is alleviated. Adversarial learning further reduces the distribution discrepancy between the target and selected source samples. In addition, we use homogeneity- and boundary-based strategies to make the pseudolabels from a preclassification map for the target domain reliable. These pixels with reliable labels are used to fine-tune the classifier of the change detection network. The experiments on five remote sensing data sets validate the efficacy of the proposed method against state-of-the-art change detection methods. In this article, the important multisource samples are selected, and the others are discarded. As future work, an important weighted scheme will be defined for multisource samples to learn a change detection model that performs well in the target domain.

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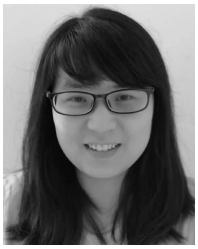
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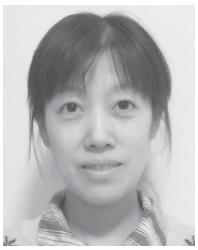
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